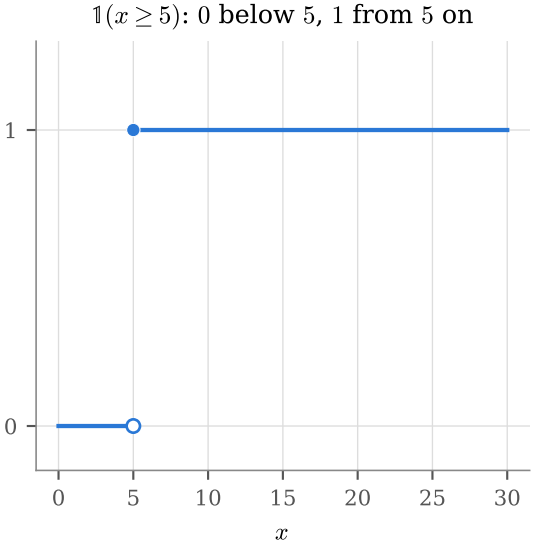
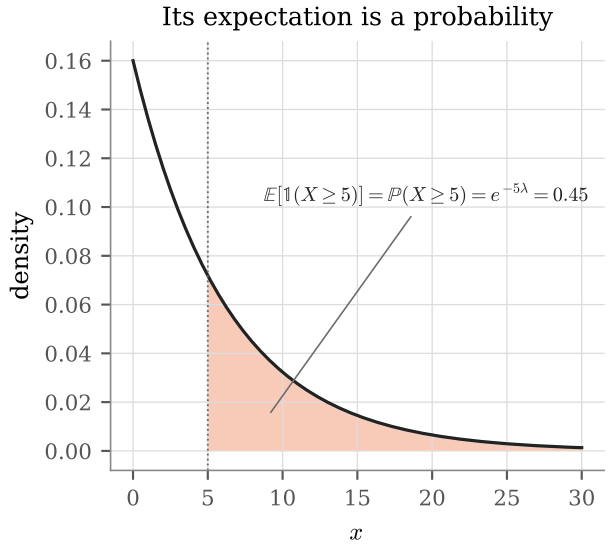

$\mathbb{1}(x \geq 5)$: 0 below 5, 1 from 5 on
1
0
0
5
10
15
20
25
30
x


Its expectation is a probability
density
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$\mathbb{E}[\mathbb{1}(X \geq 5)] = \mathbb{P}(X \geq 5) = e^{-5\lambda} = 0.45$
0
5
10
15
20
25
30
x